MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP(HONOURS)
SECOND SEMESTER PRACTICALEXAMINATION

(2024 ADMISION ONWARDS)

MG2DSCECT101 - DATA COMMUNICATION

Duration:1.5hours MaximumMarks:35

PracticalExamination–ModelQuestionPaper

Instructions:

- **Demonstrate**thegiventask.(Anyoneofthetaskslisted below)
- A**Vivasession**willfollowyourdemonstration.(Minimum5questionsfromthe Practical Module)
- **LabReport**shouldcoverallthesubunitsofthepractical module.

EvaluationCriteria:

1. **Viva** - 10 Marks
2. **Demonstration** - 15 Marks
3. **Record** - 10 Marks

Tasks:*(Chooseanyone)*

Experiments:

1. Plot and measure Amplitude and Frequency of a Sine wave
2. Plot, Amplitude and Frequency of a Square wave
3. Plot,Amplitude and Frequency of a Triangular Wave
4. Simulate any ADC circuit for various amplitudes
5. Simulate DAC for any binary weighted / ladder circuit
6. Demonstrate sampling theorem
7. Demonstrate Quantization

Optional:

8. Measure and compare the quality of the received signals, observing factors like attenuation and interference in different guided medias
9. Implement a simple delta modulationcircuit.